TSA Meeting — Slide 14 Study Note

Ch 10: Physics-Informed Fault Classifier (PINN)

Contribution C3 · Chapter 6 · SAMBP Framework

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Slide Overview and Narrative

What This Slide Shows

Slide 14 presents the **Physics-Informed Neural Network (PINN)** fault classifier, the second component of Contribution C3. The narrative is structured as a two-act argument:

Act 1 — The problem with pure ML: A standard deep neural network trained on synchronous-machine fault data misclassifies 31% of IBR-modified fault signatures. High-IBR conditions ($k_{\text{ibr}} > 0.6$) reduce baseline accuracy to 64.2%, because the network has no physical constraint to anchor it when the operating point moves outside its training distribution.

Act 2 — The physics fix: Embedding KCL as a penalty residual in the training loss forces the classifier to remain physically consistent. The physics weight $\lambda_{\text{phys}} = 0.3$ balances empirical data fit against physical law. The result: overall accuracy rises from 81.4% to 96.1%, and high-IBR accuracy jumps from 64.2% to 93.8% — a 46% relative gain precisely where pure ML was most unreliable.

Slide Key Data Points

Metric	Baseline (pure ML)	PINN
Overall accuracy	81.4%	96.1%
High-k_{ibr} (> 0.6)	64.2%	93.8%
AG fault recall	77.3%	97.4%
3ϕ fault recall	89.1%	98.2%
Physics consistency	51.2%	99.7%

Engineering Challenge: Why Pure ML Fails for IBR Faults

The Core Problem

Legacy fault classification treats the problem as purely statistical: feed in measured current features, output a class label. This works when training and test distributions match. But IBRs fundamentally change the fault physics:

- Fault current magnitude is capped at 1.1–2.0 pu by fast current limiters
- Sequence component ratios are determined by firmware, not physical reactance
- The same fault type produces dramatically different current signatures at $k_{\text{ibr}} = 0.1$ vs $k_{\text{ibr}} = 0.7$

A network trained on synchronous-machine data encounters IBR-modified signatures as *distribution shift* — it has never learned that $k_{\rm ibr}$ changes the fundamental physics of the measurement. Without a physics anchor, the network extrapolates in the wrong direction. At $k_{\rm ibr} > 0.6$, baseline accuracy collapses to 64.2%.

The three classification errors the baseline commits most often:

1. **Structural MISS** ($k_{\rm ibr} < 0.06$): Low-IBR networks still have synchronous machines; the network sees reduced fault current and misclassifies a real fault as no-fault.
2. **Z1 confusion** ($0.10 \le k_{\rm ibr} \le 0.80$): Distance-relay reach is IBR-dependent; the network predicts 87L trips when Z1 is correct.
3. **EXT false alarm** ($\alpha > 1.0$): Large normalised fault location values trigger erroneous EXTERNAL classifications.

PINN Architecture

Network Specification (TR-46)

Input dimension	4 features: $k_{\rm ibr}$, α (norm. fault location), $R_{\rm arc}$, fault type
Hidden layers	3 layers $\times$ 64 neurons, ReLU activation
Output layer	Softmax over 4 classes: MISS, 87L-only, Z1+87L, EXT
Parameters	12 676 trainable parameters
Training set	$N = 2000$ synthetic MC samples (TR-37 engine)
Measurement noise	$\sigma_k = 0.005$ pu, $\sigma_\alpha = 0.02$

The Three Physics Constraints

The PINN embeds three hard physics rules derived directly from the SAMBP protection model thresholds (TR-35):

ID	Condition	Required output	Physics source
C1	$k_{\rm ibr} < 0.06$	$P(\text{MISS}) \ge 0.95$	TR-37 MISS threshold
C2	$k_{\rm ibr} \ge 0.10$ and $\alpha \le 0.80$	$P(\text{Z1}) + P(\text{87L}) \ge 0.95$	Z1 quad reach
C3	$\alpha > 1.0$	$P(\text{EXT}) \ge 0.95$	Relay directional

These constraints directly encode the relay trip/no-trip logic — the network cannot predict Z1 for a structurally impossible IBR level, or predict 87L for a fault that relay geometry places outside reach.

The Physics-Informed Loss Function

PINN Loss: Two-Term Composite

$$\mathcal{L} \;=\; \underbrace{\mathcal{L}_{\rm data}}_{\text{cross-entropy}} + \lambda_{\rm phys} \underbrace{\left\| \mathbf{K}\hat{\mathbf{V}} - \hat{\mathbf{I}} \right\|^2}_{\text{KCL residual}}$$

$\mathcal{L}_{\text{data}}$	Standard cross-entropy classification loss
$\mathbf{K}$	Network admittance matrix (from EKF digital twin state, TR-43)
$\hat{\mathbf{V}}, \hat{\mathbf{I}}$	Estimated nodal voltages and currents from IED measurements
$\lambda_{\text{phys}} = 0.3$	Physics weighting hyperparameter (tuned by validation)

Interpretation: At each training step, the residual $\|\mathbf{K}\hat{\mathbf{V}} - \hat{\mathbf{I}}\|^2$ measures how much the predicted fault signature violates Kirchhoff's Current Law given the current network topology estimate. Penalising this residual forces the classifier to prefer hypotheses that are physically realisable — even for operating points it has never seen during training.

Why $\lambda_{\text{phys}} = 0.3$?

The value was chosen by sweep over $\{0.05, 0.10, 0.20, 0.30, 0.50, 1.00\}$ on a held-out validation set:

- Too small (< 0.10): physics constraint too weak; boundary violations persist.
- $\lambda = 0.30$: optimal trade-off — 99.7% physics consistency, 96.1% classification accuracy.
- Too large (> 0.50): physics dominates data loss; model underfits, accuracy degrades.

Physics Constraint Violations: Baseline vs PINN

Model	Violations	Rate	Root cause
Data-driven baseline	55	11.0%	No physics enforcement
PINN	5	1.0%	Noise pushes marginal k across boundary
Reduction		**90.9%**	

The 5 residual violations in the PINN all occur at $k_{\text{ibr}} \approx 0.032$ (measurement noise crosses the $k < 0.06$ threshold of constraint C1). This is a measurement limitation, not a modelling failure.

Feature Engineering and Importance (TR-47)

The 8-Feature Input Space (Slide) vs 4-Feature Core (TR-46)

The slide shows 8 input features: $|I_a|, |I_b|, |I_c|, \angle I_a, I_{\text{zero}}, I_{\text{neg}}, \hat{k_{\text{ibr}}}, \phi_{\text{ctrl}}$.
The TR-46 PINN uses a 4-feature reduced set: k_{ibr}, α, R_{arc}, fault type. This is the *physics-derived* representation where the raw current measurements have already been processed through the EKF digital twin to extract the physically meaningful quantities.

Permutation Feature Importance (TR-47)

Feature	**Importance**	**Physical meaning**
α (norm. fault location)	37.0 pp	Determines relay reach classification
k_{ibr} (IBR ratio)	27.1 pp	Controls current magnitude regime
R_{arc} (arc resistance)	21.3 pp	Causes Z1 underreach / miss errors
I_1/I_2 ratio	≈ 0 pp	Physics-derived, zero incremental importance
Z_m, Z_{ratio}	≈ 0 pp	Redundant given α and k_{ibr}
k_{margin}, α_m	≈ 0 pp	Deterministic functions of top-3 features

Key insight: Physics-derived features carry *zero incremental importance* because they are deterministic functions of the three raw features (k, α, R_{arc}). The PINN does not need them explicitly — the physics constraint in the loss function provides the same regularisation benefit without adding input dimensionality.

Minimum effective feature set: Studies B and D in TR-47 show that raw features only ($k + \alpha$, 2 features) achieve 85.5% accuracy — confirming the physics constraint, not the feature set, is the primary driver of PINN performance.

How to Read the Results Table

Per-Class Accuracy Comparison (TR-46 Figure 2)

Class	**Baseline**	**PINN**	**Gain driver**
MISS ($k_{\text{ibr}} < 0.06$)	85.2%	92.4%	C1 constraint penalises non-MISS predictions
87L ($0.06 \leq k < 0.10$)	81.5%	87.9%	C2 boundary enforced from below
Z1 ($0.10 \leq k \leq 0.80$)	64.2%	90.7%	Largest gain — C2 eliminates Z1/87L confusion
EXT ($\alpha > 1.0$)	93.0%	96.1%	C3 reinforces already-good baseline

The Z1 class shows the largest improvement (+26.5 pp) because it sits in the most crowded region of feature space — bounded by the 87L threshold below and the EXT threshold above. The physics constraint C2 acts as a discriminant surface that data alone cannot reliably learn.

Boundary Region Extrapolation (TR-46 Figure 3 — Best Figure)

The most striking result is in the *boundary region* $k \in [0.045, 0.075]$, where both constraints C1 and C2 are active simultaneously:

- **Baseline accuracy**: 75% (one-quarter of samples misclassified)
- **PINN accuracy**: 99% (+24 pp)

This is the critical operating regime for protection: the network must correctly distinguish a structural MISS ($k_{\text{ibr}} < 0.06$) from a valid 87L trip ($k_{\text{ibr}} \geq 0.10$) when the IBR measurement is noisy. The physics constraint removes the ambiguity that data alone cannot resolve.

Best Supporting Figure for This Slide

RECOMMENDED: TR-46/2026 Figure 3 – Protection Decision Space Plot

What it shows: A 2D scatter plot with k_{ibr} on the x-axis and α (normalised fault location) on the y-axis. Four coloured regions show the protection outcome zones: MISS (red), 87L-only (blue), Z1+87L (orange), EXT (grey). Overlaid scatter points show PINN predictions (open circles) vs baseline predictions (filled circles). The boundary region $k \in [0.045, 0.075]$ is highlighted. A key annotation reads "PINN: 99% / Base: 75%" in the boundary zone.

Why it is the best figure: The decision space visualisation is the single most powerful communication tool for this slide because:

1. It shows *where* the physics constraints live (the coloured zone boundaries)
2. It shows *why* the PINN wins (points cluster correctly within zones)
3. It shows *when* pure ML fails (baseline scatter leaks across zone boundaries)
4. A committee member can read the 24 pp gain directly from the figure without additional explanation

Source: TR-46/2026, page 3, Figure 3. In `PDF_COLLECTION/01_SAMBP_Reports/TR-46_sambp_report.pdf`.

Secondary figures (use if time permits):

- **TR-47 Figure 1** (feature importance bar chart): Use to answer "which features matter most?"
- **TR-49 Figure 1** (ML integration pipeline): Use to answer "how does the PINN plug into the relay?"
- **TR-46 Figure 2** (per-class accuracy bar chart): Use to answer "which fault type gains most?"

Cross-Thesis Connections

Connection	Explanation
EKF Digital Twin (Slide 13, C3)	The PINN receives $\hat{k_{\text{ibr}}}$ from the EKF as an input feature. The EKF's RMSE ≤ 0.024 pu on k_{ibr} directly bounds the worst-case PINN input error; PINN was trained with $\sigma_k = 0.005$ pu noise accordingly.
Adaptive 87L (Slide 9, C1)	PINN output class C2 ("Z1+87L") triggers the adaptive 87L relay. The PINN's 97.4% AG recall feeds into the 87L TPR = 1.000 result — misclassification would reduce 87L sensitivity.
SAMBP-SyncOC (Slide 10, C2)	The PINN fault type classification (AG/BG/3ϕ/etc.) informs the SyncOC adaptive setting selection. Wrong fault type $\Rightarrow$ wrong inverse-time curve $\Rightarrow$ mis-coordination.
CUSUM Online Learning (Chapter 6, C3)	CUSUM detects fleet composition drift ($\Delta k_{\text{ibr}} > 0.05$) and triggers PINN retraining. The PINN's 99.7% physics consistency acts as a sanity check: retraining that degrades consistency below 95% is rejected.
Physics Veto (TR-49, C4)	The PINN output passes through a 3-constraint hard veto before publishing the GOOSE advisory. Any prediction violating C1–C3 is overridden by the relay-physics deterministic rule. This eliminates all residual violations in the integration pipeline.

Equations to Know for the TSA Meeting

PINN Loss Function

$$\mathcal{L} = -\sum_i \sum_c y_{ic} \log \hat{p}_{ic} \;+\; \lambda_{\text{phys}} \sum_j \left\| \mathbf{K}_j \hat{\mathbf{V}}_j - \hat{\mathbf{I}}_j \right\|^2$$

Say: "The first term is ordinary cross-entropy; the second term penalises any prediction that requires a current distribution Kirchhoff's laws say is impossible."

Physics Constraint C2 (Z1 / 87L Boundary)

$$k_{\text{ibr}} \geq 0.10 \;\text{ and }\; \alpha \leq 0.80 \implies P(\text{Z1}) + P(\text{87L}) \geq 0.95$$

Say: "When the IBR penetration is high enough and the fault is inside relay reach, the network is forced to predict a trip. This is not an ad-hoc rule — it encodes the same threshold the SAMBP relay uses for adaptive trip decisions."

KCL Admittance Residual

$$\varepsilon_{\text{KCL}} = \|\mathbf{K}\hat{\mathbf{V}} - \hat{\mathbf{I}}\|$$

Say: "This is just Kirchhoff's Current Law in matrix form. If the network predicts a fault classification that implies a current vector $\hat{\mathbf{I}}$ inconsistent with the measured voltages and the network admittance matrix, the loss goes up. The PINN learns to avoid physically impossible predictions."

Boundary Region Accuracy Gain

$$\Delta\text{Acc}_{\text{boundary}} = 99\% - 75\% = +24 \text{ pp} \quad (k \in [0.045,\ 0.075])$$

Say: "The 24 pp gain is largest exactly where protection is most critical — the threshold between a structural miss and a valid relay trip."

Speaker Script

What to Say at the TSA Meeting

Opening (15 seconds): "Chapter 10 asks: can a neural network classify IBR fault types reliably — even when the operating point is far outside its training distribution? The short answer is no, not without physics. The baseline deep network misclassifies 31% of IBR-modified signatures because it has no knowledge of what is physically possible."

The fix (20 seconds): "We embed Kirchhoff's Current Law directly into the training loss as a KCL residual penalty, weighted at $\lambda = 0.3$. The network is penalised every time it predicts a fault class that would require a physically impossible current distribution. Three hard constraints encode the relay threshold logic from the SAMBP protection model."

Results (25 seconds): "Overall accuracy rises from 81.4 to 96.1%. But the headline result is the high-IBR accuracy: from 64.2 to 93.8% — a 46% relative gain. And in the critical boundary region where the MISS/trip decision is hardest, accuracy goes from 75 to 99%. Physics constraint violations drop from 11% to 1% — the residual 1% is measurement noise crossing the threshold, not a model error."

Integration note (15 seconds): "The PINN does not operate standalone. Its output passes through a hard physics veto before any GOOSE advisory is published. And the inputs include the IBR penetration ratio estimated by the EKF digital twin from the previous slide — the two components of C3 are directly coupled."

Closing link (10 seconds): "This completes Contribution C3: a non-blocking digital twin that tracks network state, feeding into a physics-informed classifier that makes reliable fault decisions even as the fleet composition drifts — without ever delaying a primary relay trip."

Anticipated Committee Questions and Answers

Q1: Why not just retrain the pure ML model on IBR-modified data?

Answer: Retraining addresses today's fleet composition, but the IBR ratio k_{ibr} is continuously changing as new converters are commissioned. A retrained network degrades as soon as the operating point shifts. The PINN's physics constraint is *topology-invariant* — KCL holds regardless of k_{ibr}. The CUSUM-triggered online learning (Chapter 6) handles slow drift; the physics constraint handles instantaneous generalisation.

Q2: The KCL residual uses K — the network admittance matrix. How do you get K in real time?

Answer: The EKF digital twin (TR-43/44) continuously estimates Z_{line} and Y_{shunt} from IED phasor measurements. These estimates are written to the DT library look-up table that the PINN accesses at inference time (TR-49 Figure 1, latency: 0.107 ms). The EKF's RMSE ≤ 0.024 pu on impedance bounds the worst-case $\mathbf{K}$ perturbation; this sensitivity has been tested (TR-49 Study B).

Q3: How does 96.1% compare to state-of-the-art ML fault classifiers?

Answer: Comparable pure-ML classifiers report 95–98% on synchronous-machine test sets, but degrade to 60–70% on IBR-dominated test sets (as our baseline confirms). Our 93.8% at $k_{\mathrm{ibr}} > 0.6$ is, to our knowledge, the first published result for that operating regime. The comparison is not purely about accuracy — the 99.7% physics consistency guarantee is the novel contribution, not the headline number alone.

Q4: Why use a PINN when the physics constraints C1–C3 could just be a hard post-processing veto?

Answer: The hard veto (TR-49) is also implemented as a safety layer. But the veto and the PINN serve different roles. The veto *catches* violations after inference; the PINN *prevents* them by learning to avoid physically impossible predictions in the interior of the decision space. In the boundary region ($k \in [0.045, 0.075]$), the veto cannot help because the prediction is technically inside the allowable zone — only the physics-informed loss provides the discriminant surface.

Q5: The training set is only 2000 samples. Is that enough?

Answer: Yes, because the PINN's physics constraint acts as a strong regulariser that reduces effective model complexity. Study D (TR-47) shows the minimum effective feature set is 2 features ($k+\alpha$, 85.5%), confirming the classification surface is low-dimensional. The physics constraint provides the equivalent of approximately 2 hidden layers of regularisation, so 2000 samples is adequate for a 12,676-parameter network — the Cramér–Rao argument for the EKF (TR-43) applies analogously here.

Source documents: TR-46/2026 (PINN architecture and physics constraints), TR-47/2026 (feature engineering and importance), TR-49/2026 (ML integration pipeline and physics veto).
Thesis location: Chapter 6, Contribution C3, Section on Physics-Informed Fault Classification.
Best figure: TR-46/2026, Figure 3 (decision space in (k_{ibr}, α) plane, boundary region highlighted).